

Here's a polished **research reference document write-up** combining the experiment log (CSILSTMNet) and your presentation insights. It's structured as a reference-ready section that can directly fit into a research paper's "Experimental Setup," "Results," and "Analysis" sections.

Wi-Fi-Based Human Identification and Activity Recognition using CSI-LSTM Networks

1. Introduction

This study investigates deep learning architectures for human identification and activity recognition using Wi-Fi Channel State Information (CSI) signals. CSI captures fine-grained changes in signal propagation caused by human motion, making it a powerful, non-intrusive biometric source. The goal is to design and evaluate architectures capable of processing multi-channel CSI sequences alongside metadata for accurate classification in indoor environments under both LOS and NLOS conditions.

2. Experimental Setup

2.1 Dataset Overview

The dataset used is derived from "*A Dataset for Wi-Fi-Based Human Activity Recognition in LOS and NLOS Indoor Environments*" by Alsaify et al. (2020), comprising 30 participants performing 12 unique activities. Each sample includes CSI and RSSI measurements across multiple subcarriers, with filenames encoding both subject ID and activity class.

- **Total samples:** 42,879
- **Subjects:** 30 (S1–S30)
- **Activities:** 12 (A1–A12)
- **Environments:** Lab, hallway (LOS), and barrier-separated (NLOS)
- **Features:** 99 CSI features + 12 metadata features per window

2.2 Model Design

The model **CSILSTMNet** integrates two processing streams:

- **CSI Core:** LSTM network processing temporal sequences of CSI features.
 - **Metadata Core:** Parallel LSTM block for metadata attributes.
- Outputs from both streams are concatenated and passed through dense layers for classification into 31 subject classes (for Human ID). Training employed a two-phase approach with learning rates of **0.001** and **0.0005** using **MLflow** for experiment tracking and reproducibility.

2.3 Training Configuration

Parameter	Value
Epochs	100
Optimizer	Adam
Learning Rates	0.001, 0.0005
Batch Size	128
Loss Function	Cross-Entropy
Device	Apple MPS (Metal Performance Shaders)

Each epoch averaged ~33.5 seconds. Metrics including accuracy, loss, precision, recall, and F1-score were logged via MLflow.

3. Results

3.1 Training Performance (lr = 0.001)

- **Convergence:** Training accuracy improved steadily from 62% to 99.6%.
- **Validation Accuracy:** Peaked at **98.34%** at epoch 76.
- **Loss Behavior:** Training and validation losses showed stable convergence with no overfitting beyond epoch 70.
- **Epoch Duration:** ~33.5 seconds per epoch.

Metric	Best Value (Epoch 76)
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Validation Accuracy **0.9834**

Validation Loss 0.0725

Training Accuracy 0.9956

Training Loss 0.0147

3.2 Training Performance (lr = 0.0005)

At a reduced learning rate, the network demonstrated smoother convergence and slightly better stability in validation loss. Validation accuracy reached **97.8–98.0%**, indicating marginal but consistent improvement in generalization.

4. Evaluation Metrics

4.1 Classification Report

The classification results across 31 classes yielded:

- **Macro F1-score:** 0.98
- **Precision:** 0.98
- **Recall:** 0.98
- **Support:** 8576 samples

These balanced scores across all classes confirm consistent model performance without class bias.

4.2 Confusion Matrix Insights

The confusion matrix indicates high diagonal dominance, with minor misclassifications between visually or behaviorally similar activities (e.g., A17–A18 pairs). Most classes achieved near-perfect recall (≥ 0.97).

5. Comparative Analysis with Other Architectures

Model	CSI Core	Metadata Core	Best Validation Accuracy	Learning Rate	Comments
DenseNet1D	DenseBlock1D	Bi-LSTM	0.9999	0.0005	Best overall performer
EfficientNet1DLSTM	MBConv1D	Bi-LSTM	0.9999	0.0005	Lightweight, robust generalization
CSILSTMNet	LSTM	LSTM	0.9834	0.001	Strong baseline; simpler architecture
MobileNetV3_1D_LSTM	MobileNetV3 Blocks	Bi-LSTM	0.9885	0.001	Efficient but slightly less accurate

Observation:

While CSILSTMNet achieved near-state-of-the-art accuracy, *DenseNet1D* with $lr=0.0005$ emerged as the top-performing architecture across all metrics (Accuracy, Precision, Recall,

and $F1 = 0.9999$). The lower learning rate consistently improved model generalization across all architectures.

6. Insights and Discussion

- **Learning Rate Sensitivity:** Across experiments, $lr=0.0005$ yielded more stable and higher validation accuracy compared to $lr=0.001$.
 - **Model Complexity vs. Performance:** Complex architectures like DenseNet1D outperformed simpler LSTM-based models, but CSILSTMNet remains computationally efficient with high interpretability.
 - **Generalization:** Consistently high F1 and accuracy values (>0.97) indicate strong generalization, confirming model robustness against noise and inter-subject variability.
 - **Human Identification Feasibility:** The results validate that human identification using CSI features is feasible with deep temporal models, supporting future research into non-intrusive biometric authentication.
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7. Conclusion

This work demonstrates that Wi-Fi CSI signals, when modeled with deep learning architectures, can reliably identify human subjects and recognize activities with accuracy exceeding **98%**. Among all tested architectures, **DenseNet1D** with a **learning rate of 0.0005** exhibited the highest performance ($F1 = 0.9999$). However, the **CSILSTMNet** architecture provided a balanced trade-off between model simplicity, interpretability, and performance, making it suitable for embedded or real-time applications.

Future work will focus on expanding multi-output modeling for simultaneous *Human ID* and *Activity* recognition, and exploring attention-based transformer models for enhanced temporal dependency capture.
